Assignment 1

1D Finite Element problem

Consider the structure of a body modelled as a one-dimensional elastic solid domain, as depicted in Figure 1. The cross-sectional area A and the Young's modulus E are constants along x. The structure is sustained in $x = 0$ and is subjected to the action of a horizontal body force $b(x) = c \cdot exp(\lambda x)\ N/m$ as illustrated in the figure. The equation and boundary conditions accounting for the elastic response of the structure are given by:

$$EA\frac{d^2u}{dx^2} + b(x) = 0 \tag{1}$$

$$u(0) = 0 \tag{2.a}$$

$$\frac{du}{dx}(L) = 0 \tag{2.b}$$

where $u(x)$ is the displacement. Consider the following parameters: $A = 120mm^2$, $E = 75GPa$, $L = 2m$, $c = 1 \times 10^4\ Nm^{-1}$ and $\lambda = 2$. The following is asked:

(a) Derive the weak form of equation (1) subjected to boundary conditions (2.a) and (2.b).
(b) Solve the problem using a linear displacement discretization and with different element sizes $\left(h = \frac{L}{2}, \frac{L}{4}, \cdots, \frac{L}{64}, \frac{L}{128}\right)$.
(c) Compare (by plotting) the finite element solution against the exact solution.
(d) Solve the weak form again but this time using a quadratic element. Use the element sizes suggested in (b).
(e) Compare the accuracy of the results plotting $Log\ e$ vs $Log\ h$ for the two discretizations and for the following error (*see* Appendix C. Convergence of the FEM):

$$e^u = \frac{\left\|u^{ex}(x) - u^h(x)\right\|_{L_2}}{\left\|u^{ex}(x)\right\|_{L_2}}$$

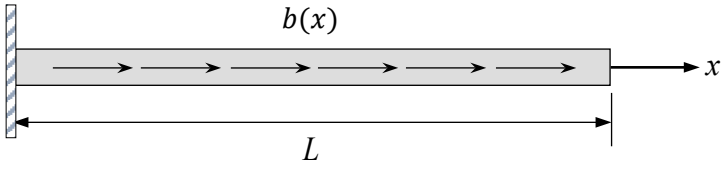


Figure 1. One dimensional solid domain.